Chapter 5 - Integral Dependence and Valuations

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Exercise 5.1
Use *Theorem 5.10*, trivial.

Exercise 5.2
Use *Theorem 5.10*, trivial.

Exercise 5.3
Because

$$\operatorname{Im}(f \otimes 1) = (\operatorname{Im} f) \otimes_A C$$

Therefore $B' \otimes_A C$ is integral over $\operatorname{Im} f \otimes_A f$, as B' is integral over $\operatorname{Im} f$.

Remark 5.1
Indeed, *Exercise 5.3* includes *Proposition 5.6(ii)* as special case, as $S^{-1}B \cong S^{-1}A \otimes_A B$.

Exercise 5.4
Following from the hinted counterexample, the answer is no.

Exercise 5.5

(i) Since $x^{-1} \in B, \exists a_{n-1}, \ldots, a_0,$ s.t.

$$x^{-n} + a_{n-1}x^{-n+1} + \ldots + a_1 x^{-1} + a_0 = 0.$$

Hence

$$x^{-1} = -a_{n-1} - a_{n-2}x - \ldots - a_1^{n-2} - a_0 x^{n-1} \in A$$

(ii) Use *Corollary 5.8*, trivial.

Exercise 5.6
Indeed. Denote the integral ring homomorphism

$$f_k : A \longrightarrow B_k$$

and we have

$$\prod_{k=1}^{n} f_k : A^n \longrightarrow \prod_{k=1}^{n} B_k$$

as an integral ring homomorphism.
(Elimination of terms to make the elements integral is easy, as we can nest those polynomials by terms.)

Exercise 5.7
Suppose $\exists x$ integral over A in $B \backslash A$, i.e.$\exists a_0, \ldots, a_{n-1} \in A,$ s.t.

$$x^n + a_{n-1}x^{n-1} + \ldots a_1 x + a_0 = 0$$

Hence

$$x(x^{n-1} + a_{n-1}x^{n-2} + \ldots + a_1) = -a_0 \in A$$

By the multiplicative closeness of $B\backslash A$,

$$x^{n-1} + a_{n-1}x^{n-2} + \ldots + a_1 \in A$$

By repeating this process, we can have

$$x + a_{n-1} \in A \Rightarrow x \in A.$$

Contradictory.

Exercise 5.8

(i) *(The hint just proved everything, there's nothing else I could say.)*

(ii) Similarly, we can think of a way to split f, g without the assisstant of algebraically closed field, as its existance requires B being integral domain:

Notice f, g, as polynomials, have finite degree. Therefore, such can be achieved by having field extension finitely many times.

□

Exercise 5.9

(I failed, but the hint proved everything again.)

Exercise 5.10